

FINAL MARK

**GIRRAWEEEN HIGH SCHOOL
MATHEMATICS EXTENSION 2
HSC ASSESSMENT TASK 1, 2017 (HSC 2018)
ANSWERS COVER SHEET**

Name: _____

QUESTION	MARK	E2	E3	E4	E5	E6	E7	E8	E9
Multiple choice	/5		✓						✓
Q6	/14		✓						✓
Q7	/21		✓						✓
Q8	/21		✓						✓
Q9	/21		✓						✓
Q10	/8		✓						✓
Q11	/20		✓						✓
TOTAL									
	/110		/110						/110

- E1 appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems.
- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.



GIRRAWEE HIGH SCHOOL

TASK 1

2017 (HSC 2018)

MATHEMATICS

EXTENSION 2

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used.
- Start each question on a separate page. Each paper must show your name.

Girraween High School Mathematics Extension 2 Test 1 2017(HSC 2018)

Multiple Choice (5 marks) Write the letter corresponding to the correct answer in your answer booklet.

1. What is the value of i^{2018} ?

(A) 1

(B) -1

(C) 0

(D) -2

2. If $x + iy = \frac{a + ib}{c + id}$ then $x^2 + y^2 = ?$

(A) $\frac{a^2 + b^2}{c^2 + d^2}$

(B) $\frac{a^2 - b^2}{c^2 + d^2}$

(C) $\frac{a^2 + b^2}{c^2 - d^2}$

(D) $\frac{a^2 - b^2}{c^2 - d^2}$

3. Multiplying a non-zero complex number by $\frac{1+i}{1-i}$ results in a rotation about the origin on an Argand diagram. What is the rotation ?

(A) clockwise by $\frac{\pi}{2}$

(B) clockwise by $\frac{\pi}{4}$

(C) anticlockwise by $\frac{\pi}{2}$

(D) anticlockwise by $\frac{\pi}{4}$

4. If $z = 2 + i$ and $w = 1 - i$, then $z\bar{w}$, in the form $x + iy$ is:

(A) $3 - i$

(B) $1 - 3i$

(C) $1 + 3i$

(D) $1 + i$

5. Let $\bar{z} = 5 + 6i$. Find $\text{Im}(3i - z)$

(A) $3i$

(B) $-3i$

(C) 3

(D) -3

Question 6 (14 marks)

Marks

(a) If $z = \frac{-1+i}{\sqrt{3}+i}$

(i) Express z in the form $x+iy$, where x and y are real. 2

(ii) By expressing $-1+i$ and $\sqrt{3}+i$ in the modulus argument form, show that

$$z = \frac{1}{\sqrt{2}} \operatorname{cis}\left(\frac{7\pi}{12}\right) \quad 3$$

(iii) Hence find the exact value of $\tan\frac{7\pi}{12}$ 3

(b) (i) Find all pairs of real numbers x and y such that $(x+iy)^2 = -5-12i$. 3

(ii) Hence solve: $z^2 - 4z + (9+12i) = 0$ 3

Question 7 (21 marks)

(a) (i) Express $\sin 5\theta$ and $\cos 5\theta$ in terms of $\cos \theta$ and $\sin \theta$. 4

(ii) Hence express $\tan 5\theta$ in terms of $\tan \theta$. 3

(b) (i) If $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ and $z^n - \frac{1}{z^n} = 2i \sin n\theta$. 4

(ii) Use De Moivre's theorem to obtain an expression for $\cos^7 \theta$ in the form

$a \cos 7\theta + b \cos 5\theta + c \cos 3\theta + d \cos \theta$. What are the values of a, b, c and d . 4

(c) $1, \omega, \omega^2$ are the roots of the equation $z^3 = 1$.

(i) Show that $1 + \omega + \omega^2 = 0$ 2

(ii) Find the cubic equation whose roots are $1, 1+\omega, 1+\omega^2$. 4

Question 8 (21 marks)

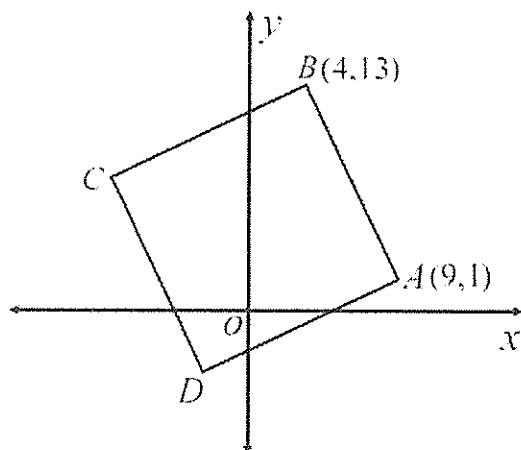
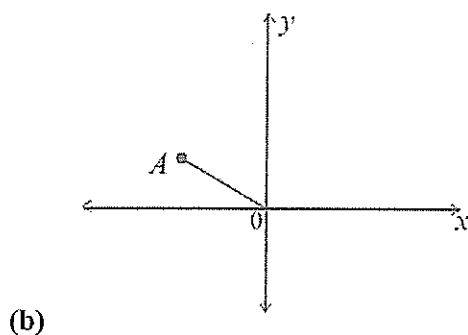
- (a) (i) Express the roots of the equation $z^5 + 1 = 0$ in modulus-argument form. 3
- (ii) Show the roots of $z^5 + 1 = 0$ in an Argand diagram. 3
- (iii) Show that $z^4 - z^3 + z^2 - z + 1 = \left(z^2 - 2\cos\frac{\pi}{5}z + 1\right)\left(z^2 - 2\cos\frac{3\pi}{5}z + 1\right)$ 4
- (iv) Hence find the value of $\cos\frac{\pi}{5} + \cos\frac{3\pi}{5}$ and $\cos\frac{\pi}{5}\cos\frac{3\pi}{5}$. 4
- (v) Form a quadratic equation with roots $\cos\frac{\pi}{5}$ and $\cos\frac{3\pi}{5}$. 3
- (vi) Hence find the exact value of $\cos\frac{\pi}{5}$ and $\cos\frac{3\pi}{5}$ in simplest surd form. 4

Question 9 (21 marks)

- (a) Sketch the following loci on separate Argand diagram.
- (i) $|z + 2 - 3i| = |z + 2 + i|$ 3
- (ii) $\arg(z - (2 + 3i)) = -\frac{\pi}{3}$ 3
- (iii) $2 < |z - 1| \leq 3$ 3
- (b) Find the locus of z and sketch on an Argand diagram.
- (i) $|z - 4i| = \operatorname{Im}(z)$ 4
- (ii) $\left|z^2 - \left(\frac{\overline{z}}{z}\right)^2\right| \geq 12$ 4
- (c) Find the locus of z , given $\frac{z-4}{z-2i}$ is a purely imaginary number. 4

Question 10 (8 marks)

- (a) $\triangle OAB$ is an isosceles triangle with $OA = OB$ and $\angle OBA = 75^\circ$. If O is the origin and A represent the complex number $-\sqrt{3} + i$, find two possible complex numbers represented by the point B , in the form $a + ib$. 4



The diagram above shows a square $ABCD$ in the complex plane. The vertices A and B represent the complex numbers $9 + i$ and $4 + 13i$ respectively. Find the complex numbers represented by

- (i) The vector AB . 2
- (ii) The vertex D 2

Question 11 (20 marks)

(a) Given the equation $\arg\left(\frac{z-2}{z+4}\right) = \frac{3\pi}{4}$.

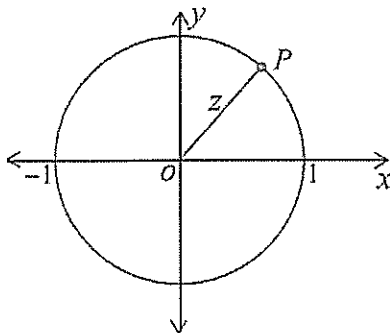
(i) Draw $\overrightarrow{z-2}$, $\overrightarrow{z+4}$, $\arg(z-2)$ and $\arg(z+4)$ in an Argand diagram and mark the

angle representing $\arg\left(\frac{z-2}{z+4}\right)$ giving reasons. 4

(ii) Find the equation of the locus of $\arg\left(\frac{z-2}{z+4}\right) = \frac{3\pi}{4}$, describe the locus and sketch in

the Argand diagram drawn in (i). All reasons must be given. 7

(b) P represents the complex number z such that $|z|=1$. Copy the diagram onto your answer sheet and prove that



(i) $\arg(z+1) = \frac{1}{2} \arg z$ 3

(ii) $\arg(z-1) = \arg(z+1) + \frac{\pi}{2}$ 3

(iii) $\left| \frac{z-1}{z+1} \right| = \tan\left(\frac{1}{2} \arg z\right)$ 3

END OF EXAMINATION

Extension 2 Task 1, 2017 (HSC 2018) - solutions

Multiple choice (5 marks)

1B 2A 3C 4C 5D

$$\begin{aligned} 1. i^{2018} &= i^{(504 \times 4) + 2} \\ &= i^2 = -1 \end{aligned}$$

$$\begin{aligned} 2. x+iy &= \frac{a+ib}{c+id} \\ x-iy &= \frac{a-ib}{c-id} \end{aligned}$$

$$\begin{aligned} (x+iy)(x-iy) &= \frac{a+ib}{c+id} \times \frac{a-ib}{c-id} \\ &= \frac{a^2+b^2}{c^2+d^2} \end{aligned}$$

$$x^2+y^2 = \frac{a^2+b^2}{c^2+d^2}$$

$$\begin{aligned} 3. \frac{1+i}{1-i} &= \frac{1+i}{1-i} \times \frac{1+i}{1+i} \\ &= \frac{(1+i)^2}{2} \\ &= \frac{1+2i-1}{2} = i \end{aligned}$$

$$4. z = 2+i, w = 1-i$$

$$\bar{w} = 1+i$$

$$z\bar{w} = (2+i)(1+i)$$

$$= 2+2i+i+i^2$$

$$= 2+3i-1$$

$$= 1+3i$$

$$5. z = 5+6i$$

$$3i-z = 3i-5-6i$$

$$= -5-3i$$

$$\operatorname{Im}(3i-z) = -3$$

Question 6 (14 marks)

$$\begin{aligned}
 (a)(i) Z &= \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} \\
 &= \frac{-\sqrt{3}+i+i\sqrt{3}-i^2}{3-i^2} \\
 &= \frac{-\sqrt{3}+i+i\sqrt{3}+1}{4} \\
 &= \frac{1-\sqrt{3}}{4} + \frac{i(1+\sqrt{3})}{4} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \frac{-1+i}{\sqrt{3}+i} \\
 |Z| &= \sqrt{1^2+1^2} = \sqrt{2} \\
 \tan \alpha &= 1, \alpha = \frac{\pi}{4} \\
 \theta &= \pi - \frac{\pi}{4} = \frac{3\pi}{4} \\
 -1+i &= \sqrt{2} \operatorname{cis} \frac{3\pi}{4} \\
 Z &= \frac{\sqrt{3}+i}{\sqrt{3}+i} = Z \\
 |Z| &= \sqrt{3+1} = 2 \\
 \tan \alpha &= \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6} \\
 \theta &= \alpha = \frac{\pi}{6} \\
 \sqrt{3}+i &= 2 \operatorname{cis} \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 \frac{-1+i}{\sqrt{3}+i} &= \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{3\pi}{4} - \frac{\pi}{6} \right) \\
 &= \frac{1}{\sqrt{2}} \operatorname{cis} \left(\frac{7\pi}{12} \right) \quad (3)
 \end{aligned}$$

(iii)

$$\frac{1-\sqrt{3}}{4} + \frac{i(1+\sqrt{3})}{4} = \frac{1}{\sqrt{2}} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right)$$

Equating real and imaginary parts

$$\frac{1}{\sqrt{2}} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4}$$

$$\frac{1}{\sqrt{2}} \sin \frac{7\pi}{12} = \frac{1+\sqrt{3}}{4}$$

$$\sin \frac{7\pi}{12} = \frac{\sqrt{2}(1+\sqrt{3})}{4}$$

$$\cos \frac{7\pi}{12} = \frac{\sqrt{2}(1-\sqrt{3})}{4} \quad (3)$$

$$\begin{aligned}
 \tan \frac{7\pi}{12} &= \frac{\sqrt{2}(1+\sqrt{3})}{4} \times \frac{4}{\sqrt{2}(1-\sqrt{3})} \\
 &= \frac{1+\sqrt{3}}{1-\sqrt{3}}
 \end{aligned}$$

$$(b)(i) (x+iy)^2 = -5-12i$$

$$x^2 + 2ixy - y^2 = -5-12i$$

$$x^2 - y^2 = -5 \quad (1)$$

$$2xy = -12$$

$$xy = -6 \quad (2)$$

$$(2) \Rightarrow y = \frac{-6}{x}$$

Substitute in (1)

$$x^2 - \frac{36}{x^2} = -5$$

$$x^4 - 36 = -5x^2$$

$$x^4 + 5x^2 - 36 = 0$$

$$\text{Let } m = x^2$$

$$m^2 + 5m - 36 = 0$$

$$(m+9)(m-4) = 0$$

$$m = -9 \text{ or } m = 4$$

$$x^2 = -9 \text{ or } x^2 = 4$$

$$x = \pm 2 \text{ (}\because x \text{ is real)}$$

$$x = 2, y = \frac{-6}{2} = -3$$

$$x = -2, y = \frac{-6}{-2} = 3$$

$\therefore$ the roots are

$$2-3i, -2+3i$$

$$= \pm(2-3i) \quad (3)$$

$$(ii) z^2 - 4z + (9+12i) = 0$$

$$z = \frac{4 \pm \sqrt{16 - 4(9+12i)}}{2}$$

$$= \frac{4 \pm \sqrt{-20-48i}}{2}$$

$$= \frac{4 \pm \sqrt{4(-5-12i)}}{2}$$

$$= \frac{4 \pm 2\sqrt{-5-12i}}{2}$$

$$= \frac{2(2 \pm \sqrt{-5-12i})}{2} \text{ page 3}$$

$$= 2 \pm \sqrt{-5-12i}$$

$$= 2 \pm 2-3i$$

$$= 2+2-3i \text{ or } 2-(2-3i)$$

$$= 4-3i \text{ or } 2-2+3i$$

$$= \underline{4-3i} \text{ or } \underline{3i} \quad (3)$$

Question 7 (21 marks)

$$(a)(i) (\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta \quad (1)$$

by De Moivre's Theorem

Using binomial theorem,

$$(\cos \theta + i \sin \theta)^5$$

$$= \cos^5 \theta + {}^5C_1 \cos^4 \theta \times i \sin \theta$$

$$+ {}^5C_2 \cos^3 \theta (i \sin \theta)^2$$

$$+ {}^5C_3 \cos^2 \theta \times i^3 \sin^3 \theta$$

$$+ {}^5C_4 \cos \theta i^4 \sin^4 \theta + i^5 \sin^5 \theta$$

$$= \cos^5 \theta + 5i \cos^4 \theta \sin \theta$$

$$- 10 \cos^3 \theta \sin^2 \theta - 10i \cos^2 \theta \sin^3 \theta$$

$$+ 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

(2)

Equating real and imaginary parts of (1) and (2)

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \quad \text{Page 4}$$

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \quad (4)$$

$$(ii) \tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

Divide numerator and denominator by $\cos^5 \theta$

$$\frac{5 \frac{\sin \theta}{\cos \theta} - 10 \frac{\sin^3 \theta}{\cos^3 \theta} + \frac{\sin^5 \theta}{\cos^5 \theta}}{1 - 10 \frac{\sin^2 \theta}{\cos^2 \theta} + 5 \frac{\sin^4 \theta}{\cos^4 \theta}} \quad (3)$$

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$(b)(i) z = \cos \theta + i \sin \theta$$

$$z^n = (\cos \theta + i \sin \theta)^n$$

$$= \cos n\theta + i \sin n\theta \quad \text{by De Moivre's theorem} \quad (1)$$

$$\frac{1}{z^n} = z^{-n} = (\cos \theta + i \sin \theta)^{-n}$$

$$= \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta \quad \left(\begin{array}{l} \because \sin(-\theta) = -\sin \theta \\ \cos(-\theta) = \cos \theta \end{array} \right)$$

$$\quad \quad \quad (2)$$

Adding (1) and (2)

$$\begin{aligned} z^n + \frac{1}{z^n} &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned}$$

① - ②

page 5

$$z^n - \frac{1}{z^n} = \cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta$$

$$= 2i \sin n\theta \quad (4)$$

$$(ii) (2 \cos \theta)^7 = \left(z + \frac{1}{z}\right)^7$$

$$= z^7 + 7C_1 z^6 \times \frac{1}{z} + 7C_2 z^5 \times \frac{1}{z^2} + 7C_3 z^4 \times \frac{1}{z^3} + 7C_4 z^3 \times \frac{1}{z^4}$$

$$+ 7C_5 z^2 \times \frac{1}{z^5} + 7C_6 z \times \frac{1}{z^6} + 7C_7 \frac{1}{z^7}$$

$$= z^7 + 7z^5 + 21z^3 + 35z + 35 \times \frac{1}{z} + 21 \times \frac{1}{z^3} + 7 \times \frac{1}{z^5} + \frac{1}{z^7}$$

$$= \left(z^7 + \frac{1}{z^7}\right) + 7 \left(z^5 + \frac{1}{z^5}\right) + 21 \left(z^3 + \frac{1}{z^3}\right) + 35 \left(z + \frac{1}{z}\right)$$

$$= 2 \cos 7\theta + 7 \times 2 \cos 5\theta + 21 \times 2 \cos 3\theta + 35 \times 2 \cos \theta$$

$$= 2 \cos 7\theta + 14 \cos 5\theta + 42 \cos 3\theta + 70 \cos \theta$$

$$2^7 \cos^7 \theta = 2 \cos 7\theta + 14 \cos 5\theta + 42 \cos 3\theta + 70 \cos \theta$$

$$2^6 \cos^7 \theta = \cos 7\theta + 7 \cos 5\theta + 21 \cos 3\theta + 35 \cos \theta$$

$$\cos^7 \theta = \frac{1}{64} \cos 7\theta + \frac{7}{64} \cos 5\theta + \frac{21}{64} \cos 3\theta + \frac{35}{64} \cos \theta$$

$$a = \frac{1}{64}, \quad b = \frac{7}{64}, \quad c = \frac{21}{64}, \quad d = \frac{35}{64} \quad (4)$$

$$(C) (i) z^3 - 1 = 0$$

$$1 + \omega + \omega^2 = -\frac{\text{coefficient of } z^2}{\text{coefficient of } -z^3} = -\frac{0}{1} = 0$$

$$(ii) \text{ sum of roots } 1 + 1 + \omega + 1 + \omega^2 = 2 \quad (2)$$

Sum of roots taken two at a time

$$\begin{aligned}
 & 1 + \omega + 1 + \omega^2 + (1 + \omega)(1 + \omega^2) \\
 &= 1 + \omega + 1 + \omega^2 + 1 + \omega^2 + \omega + \omega^3 \\
 &= \cancel{1} + \cancel{\omega} + 1 + \cancel{\omega^2} + \cancel{1} + \cancel{\omega^2} + \cancel{\omega} + 1 = 2
 \end{aligned}$$

Product of the roots

$$\begin{aligned}
 (1 + \omega)(1 + \omega^2) &= 1 + \omega^2 + \omega + \omega^3 \\
 &= 1 + \omega^2 + \omega + 1 = 1
 \end{aligned}$$

(4)

$$\therefore \text{the equation is } z^3 - 2z^2 + 2z - 1 = 0$$

or

$$\underline{\underline{2x^3 - 2x^2 + 2x - 1 = 0}}$$

Question 8 (24 marks)

(a)(i) $z^5 + 1 = 0$

$$z^5 = -1 = \cos \pi + i \sin \pi$$

$$= \cos(2k\pi + \pi) + i \sin(2k\pi + \pi) \quad k = 0, 1, 2, \dots$$

The five fifth roots of -1 are given by

$$z = [\cos(2k+1)\pi + i \sin(2k+1)\pi]^{\frac{1}{5}}$$

$$= \cos \frac{(2k+1)\pi}{5} + i \sin \frac{(2k+1)\pi}{5} \quad k = 0, 1, 2, 3, 4$$

by De Moivre's theorem

When $k=0$

$$z_1 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$$

$$k=1$$

$$z_2 = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$$

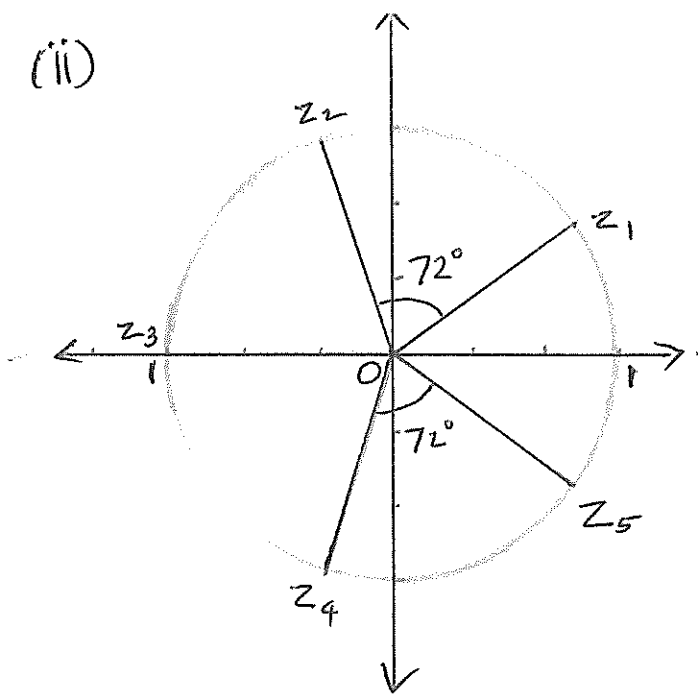
$$k=2, \quad z_3 = \cos \pi = -1$$

$$k=3, \quad z_4 = \cos \frac{7\pi}{5}$$

$$k=4, \quad z_5 = \cos \frac{9\pi}{5}$$

(3)

(ii)



3

(iii) From the above figure

$$z_5 = \bar{z}_1 \text{ and } z_4 = \bar{z}_2$$

$$z_1 + z_5 = z_1 + \bar{z}_1 = 2 \cos \frac{\pi}{5}$$

$$z_2 + z_4 = z_2 + \bar{z}_2 = 2 \cos \frac{3\pi}{5}$$

$$z_1 z_5 = z_1 \bar{z}_1 = |z_1|^2 = 1$$

$$z_2 z_4 = z_2 \bar{z}_2 = |z_2|^2 = 1$$

$$z^5 + 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$$

$$= (z + 1)(z - z_1)(z - z_5)(z - z_2)(z - z_4)$$

$$= (z + 1)(z^2 - z(z_1 + z_5) + z_1 z_5)(z^2 - z(z_2 + z_4) + z_2 z_4)$$

$$= (z + 1)\left(z^2 - 2 \cos \frac{\pi}{5} z + 1\right)\left(z^2 - 2 \cos \frac{3\pi}{5} z + 1\right) \quad \text{①}$$

$$z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1) \quad \text{②}$$

From (1) and (2) we get

$$z^4 - z^3 + z^2 - z + 1 = (z^2 - 2\cos\frac{\pi}{5}z + 1)(z^2 - 2\cos\frac{3\pi}{5}z + 1) \quad (4)$$

$$(iv) z^4 - z^3 + z^2 - z + 1 = z^4 - 2\cos\frac{3\pi}{5}z^3 + z^2 - 2\cos\frac{\pi}{5}z^3 + 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5}z^2 - 2\cos\frac{\pi}{5}z + z^2 - 2\cos\frac{3\pi}{5}z + 1$$

Equating coefficients of z^2

$$1 = 1 + 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5} + 1$$

$$1 = 2 + 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5}$$

$$-1 = 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5}$$

$$\cos\frac{\pi}{5}\cos\frac{3\pi}{5} = -\frac{1}{4}$$

Equating coefficients of z^3

$$-1 = -2\cos\frac{3\pi}{5} - 2\cos\frac{\pi}{5}$$

$$-1 = -2\left(\cos\frac{3\pi}{5} + \cos\frac{\pi}{5}\right) \quad (4)$$

$$\cos\frac{3\pi}{5} + \cos\frac{\pi}{5} = \frac{1}{2}$$

(v) The quadratic equation with roots $\cos\frac{\pi}{5}$ and $\cos\frac{3\pi}{5}$ is given by $x^2 - \frac{1}{2}x - \frac{1}{4} = 0$

$$\underline{\underline{4x^2 - 2x - 1 = 0}} \quad (3)$$

$$(vi) 4x^2 - 2x - 1 = 0$$

$$x = \frac{2 \pm \sqrt{4 - 4 \times 4 \times -1}}{8}$$

$$= \frac{2 \pm 2\sqrt{5}}{8} = \frac{1 \pm \sqrt{5}}{4}$$

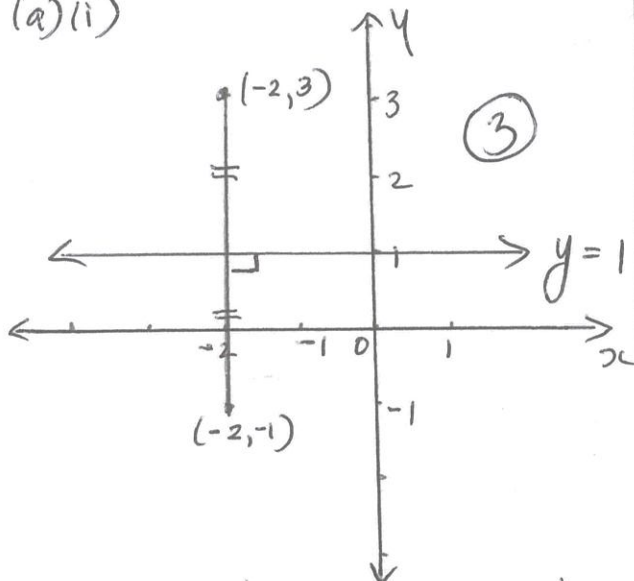
$\cos \frac{\pi}{5}$ is positive.

$\cos \frac{3\pi}{5}$ is negative

$$\therefore \cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4} \quad \text{and} \quad \cos \frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$$

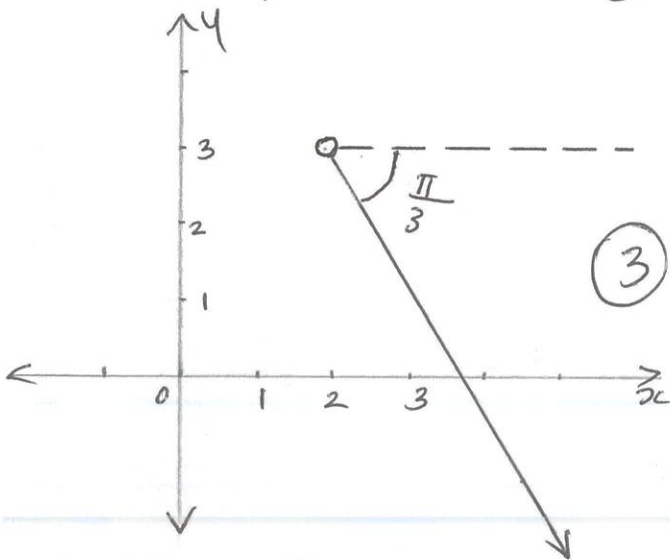
Question 9 (21 marks)

(a)(i)

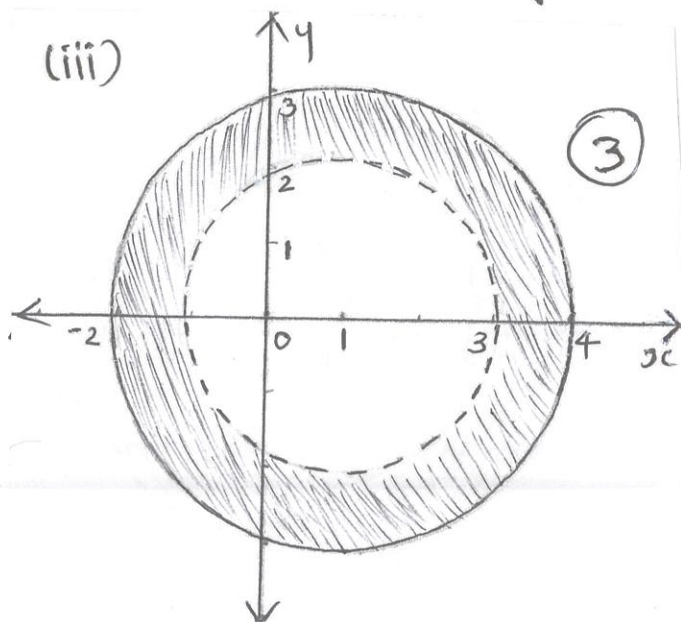


$$|z+2-3i| = |z+2+i|$$

(ii) $\arg(z - (2+3i)) = -\frac{\pi}{3}$



(iii)



(b) $|z-4i| = \operatorname{Im}(z)$

$$|x+iy-4i| = y$$

$$|x+i(y-4)| = y$$

$$\sqrt{x^2 + (y-4)^2} = y$$

$$x^2 + (y-4)^2 = y^2$$

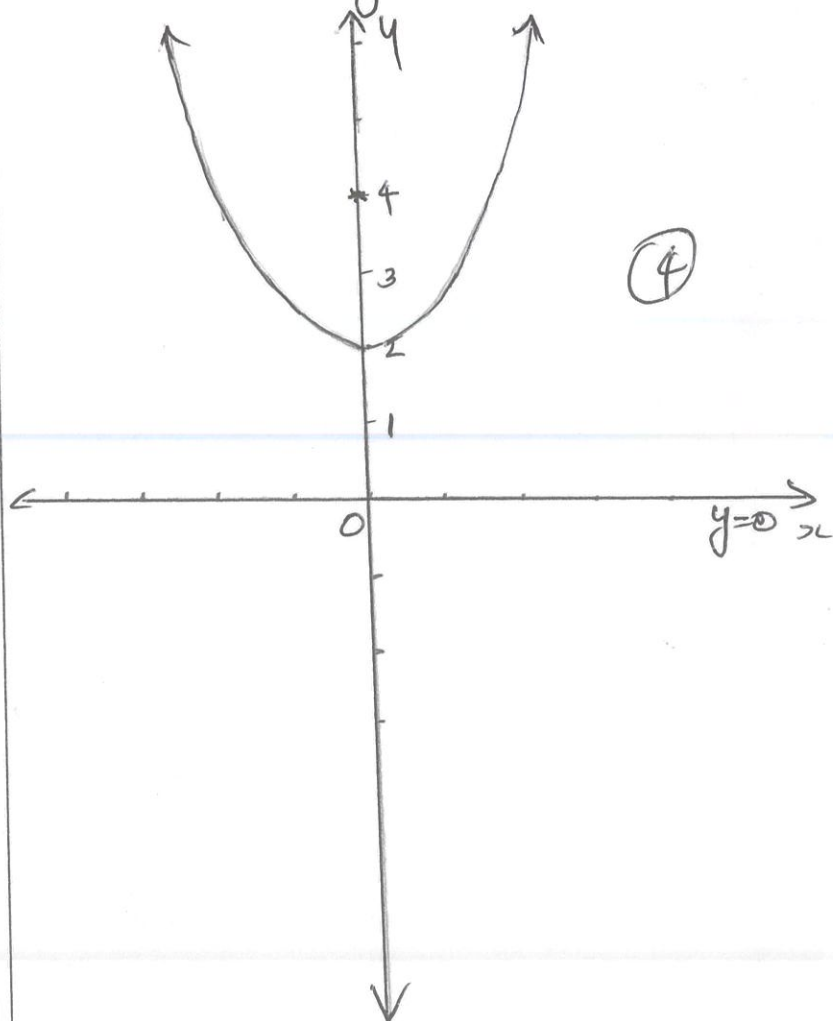
$$x^2 + y^2 - 8y + 16 = y^2$$

$$x^2 = 8y - 16$$

$$= 8(y-2)$$

Vertex (0, 2) Focus (0, 4)

Directrix $y=0$



$$(ii) |z^2 - (\bar{z})^2| \geq 12$$

$$|(x+iy)^2 - (x-iy)^2| \geq 12$$

$$|(x+iy + x-iy)(x+iy - x+iy)| \geq 12$$

$$|(2x)(2iy)| \geq 12$$

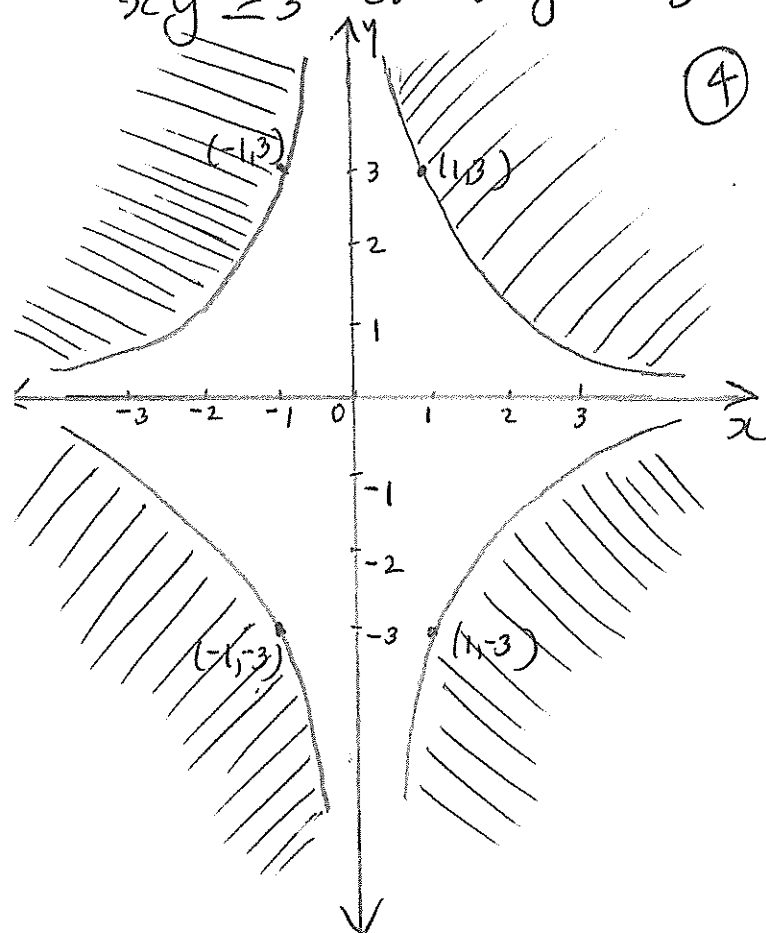
$$|4ixy| \geq 12$$

$$4|xy| \geq 12$$

$$|xy| \geq 3$$

$$xy \geq 3 \quad \text{or} \quad -xy \geq 3$$

$$xy \geq 3 \quad \text{or} \quad xy \leq -3$$



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$$(iii) \text{ Let } z = x+iy$$

$$\frac{z-4}{z-2i} = \frac{x+iy-4}{x+iy-2i}$$

$$= \frac{x-4+iy}{x+iy-2i} \times \frac{x-i(y-2)}{x-i(y-2)}$$

$$= \frac{x(x-4) - i(x-4)(y-2) + ixy - i^2y(y-2)}{x^2 - i^2(y-2)^2}$$

$$= \frac{x(x-4) + y(y-2) + ixy - i(x-4)(y-2)}{x^2 + (y-2)^2}$$

$$\operatorname{Re}\left(\frac{z-4}{z-2i}\right) = \frac{x(x-4) + y(y-2)}{x^2 + (y-2)^2} = 0$$

$$x(x-4) + y(y-2) = 0$$

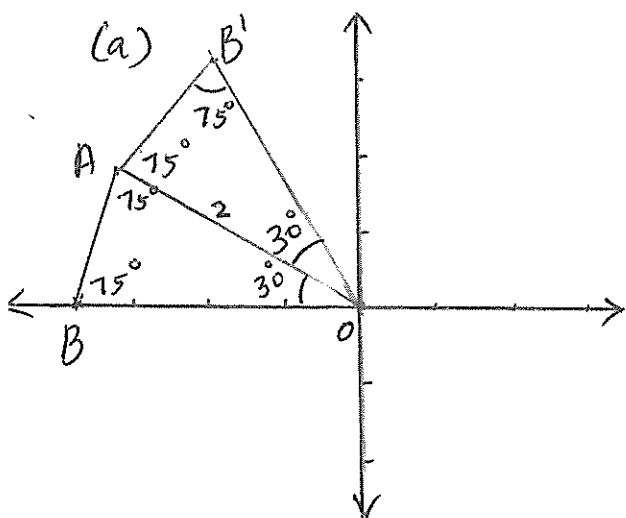
$$x^2 - 4x + y^2 - 2y = 0$$

$$x^2 - 4x + 4 + y^2 - 2y + 1 = 0$$

$$(x-2)^2 + (y-1)^2 = 5$$

(4)

Question 10 (8 marks)



$$z = \sqrt{3} + i$$

$$|z| = \sqrt{3+1} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \quad \alpha = \frac{\pi}{6}$$

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\arg z = \frac{5\pi}{6}$$

$$\angle BOA = 30^\circ \text{ and}$$

$$\angle AOB' = 30^\circ$$

$$\vec{OB} = -2$$

$$\vec{OB'} = 2 \operatorname{cis} \frac{2\pi}{3}$$

$$= 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= 2 \left(-\frac{1}{2} + i \times \frac{\sqrt{3}}{2} \right)$$

$$= \underline{\underline{-1 + i\sqrt{3}}}$$

$$B = -2 + 0i$$

$$B' = -1 + i\sqrt{3}$$

(b)

page 12

$$\begin{aligned} \text{(i)} \quad \vec{AB} &= \vec{OB} - \vec{OA} \\ &= 4 + 13i - 9 - i \\ &= -5 + 12i \end{aligned}$$

(2)

$$\begin{aligned} \text{(ii)} \quad \vec{AB} &= i(-5 + 12i) \\ &= -5i + 12i^2 = -12 - 5i \end{aligned}$$

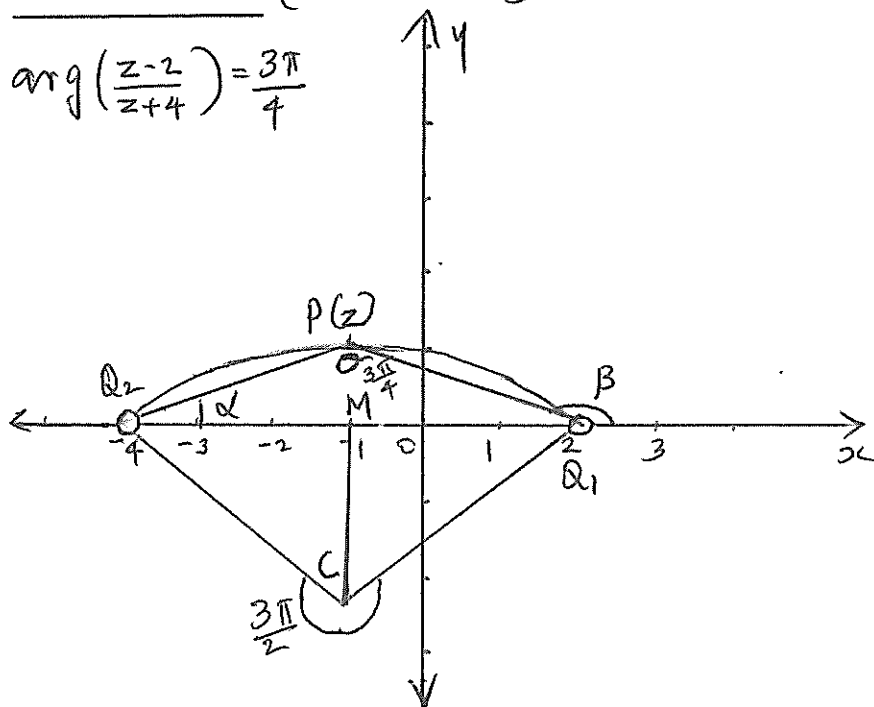
$$\begin{aligned} \vec{OD} &= \vec{OA} + \vec{AB} \\ &= (9 + i) + (-12 - 5i) \\ &= -3 - 4i \end{aligned}$$

(2)

$$\underline{\underline{D = (-3 - 4i)}}$$

Question 11 (20 marks)

$$\arg \left(\frac{z-2}{z+4} \right) = \frac{3\pi}{4}$$



$$\arg(z-2) = \angle PQ_1X = \beta$$

$$\arg(z+4) = \angle PQ_2X = \alpha$$

$$\alpha + \theta = \beta \text{ (exterior angle of } \triangle Q_2PQ_1)$$

$$\theta = \beta - \alpha$$

$$= \arg(z-2) - \arg(z+4)$$

(4)

$$= \arg\left(\frac{z-2}{z+4}\right) = \frac{3\pi}{4}$$

(ii) Let C be the centre of the circle. Draw $CM \perp Q_1Q_2$.
 M is the mid point of Q_1Q_2 since the perpendicular from the centre of a circle to a chord bisects the chord. $M = (-1, 0)$

Reflex $\angle Q_2CQ_1 = \frac{3\pi}{2}$ (angle at the centre is twice angle at the circumference standing on the same arc)

$\triangle Q_2CQ_1$ is isosceles ($Q_2C = Q_1C = \text{radii}$)

$$\angle Q_2CQ_1 = 2\pi - \frac{3\pi}{2} = \frac{\pi}{2} \text{ (angles at a point)}$$

$$\begin{aligned} \angle CQ_2Q_1 &= \frac{\pi - \frac{\pi}{2}}{2} \text{ (angles opposite equal sides in an isosceles } \triangle, \text{ angle sum of } \triangle) \\ &= \frac{\pi}{4} \end{aligned}$$

(7)

In $\triangle Q_2CM$

$$\tan \frac{\pi}{4} = \frac{CM}{Q_2M} = \frac{CM}{3}$$

$$\frac{CM}{3} = 1; CM = 3$$

Centre $(-1, -3)$

Apply Pythagoras' theorem in $\triangle Q_2CM$

$$Q_2C = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

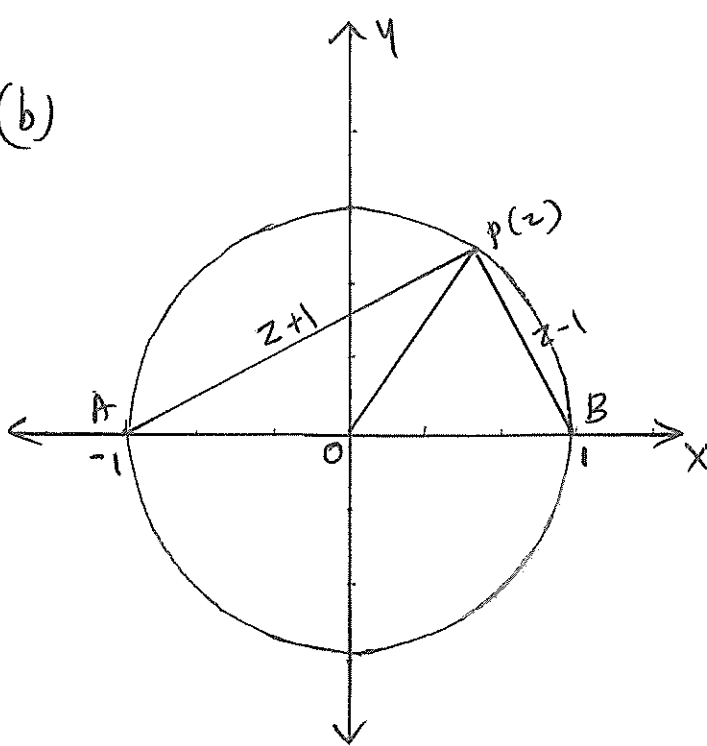
locus is the minor arc of the circle

$$(x+1)^2 + (y+3)^2 = 18, y > 0$$

excluding the points

$$(-4, 0) \text{ and } (2, 0)$$

(b)



(i) $\angle PAO + \angle APO = \angle POB$ (exterior $\angle$ of $\triangle AOP$)

$$\arg(z+1) + \arg(z+1) = \arg z \quad \left(\because \triangle AOP \text{ is isosceles} \right.$$

$OA = OP = 1$, angles opposite equal sides in isosceles $\triangle$)

$$2 \arg(z+1) = \arg z$$

$$\arg(z+1) = \frac{1}{2} \arg z$$

(3)

(ii) $\angle APB = 90^\circ$ (angle in a semicircle)

$$\angle PAB + \angle APB = \angle PBX$$

$$\arg(z+1) + \frac{\pi}{2} = \arg(z-1)$$

$$\arg(z-1) = \arg(z+1) + \frac{\pi}{2}$$

(3)

$$(iii) \tan \angle PAB = \frac{PB}{PA} = \frac{|z-1|}{|z+1|} = \left| \frac{z-1}{z+1} \right|$$

$$\tan(\arg(z+1)) = \left| \frac{z-1}{z+1} \right| = \tan\left(\frac{1}{2} \arg z\right)$$

From (i)

(3)